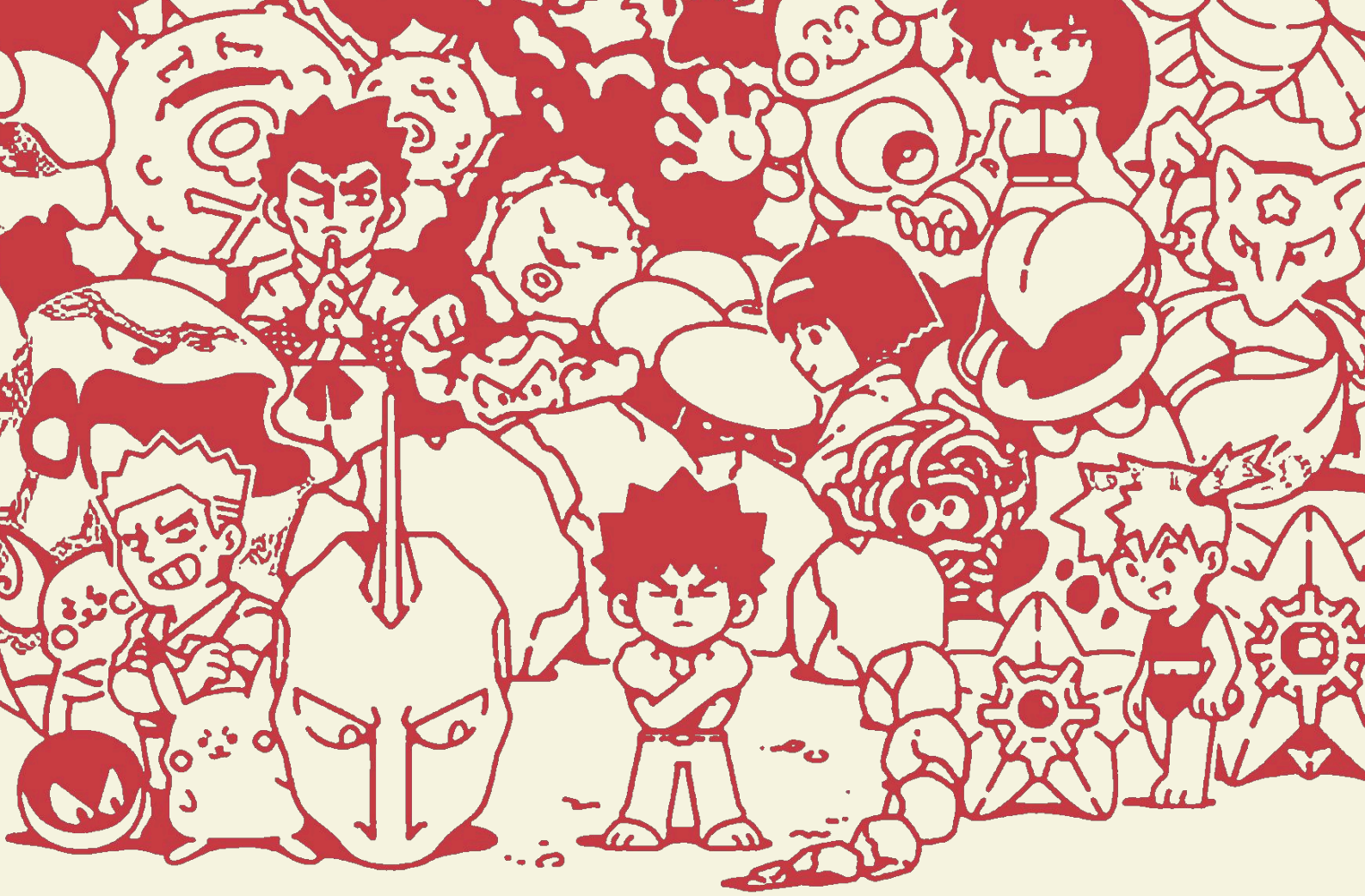
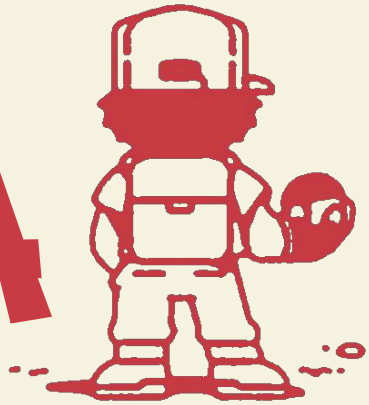




GOTTA CODE EM ALL

ODT CAPSTONE PROJECT

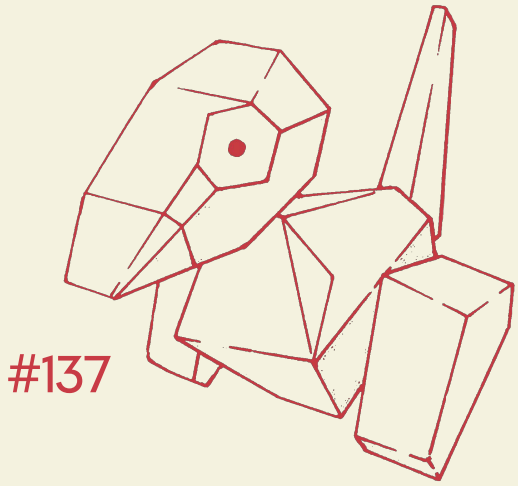
Sachit Patnaik
Tanay Dhavate

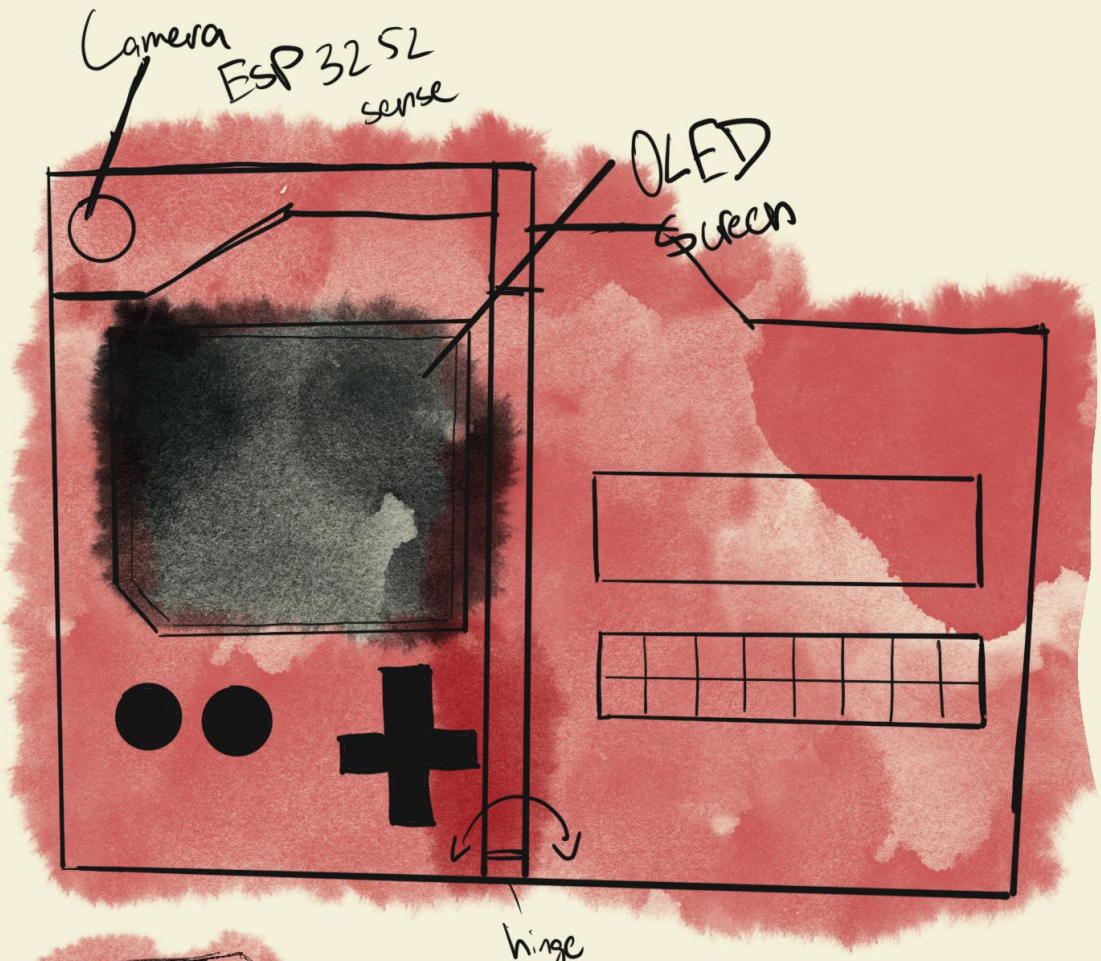
BITS Design
2025-29

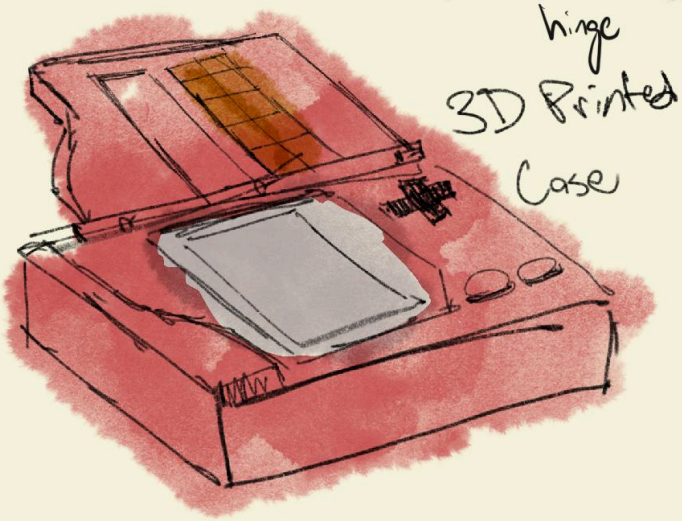
CONCEPT

A Pokédex is a fictional electronic device from the Pokémon anime, games, and wider franchise that serves as a digital encyclopaedia for Pokémon trainers. It is used to identify and record information about different Pokémon encountered in the world, providing details such as a Pokémon's name, type, physical characteristics, and a short descriptive entry unique to each creature. Phe Pokédex represents the spirit of exploration and discovery that sits at the heart of the Pokémon universe, making it a, in our minds a compelling inspiration for a project that brings together identification, data display, and physical interaction.

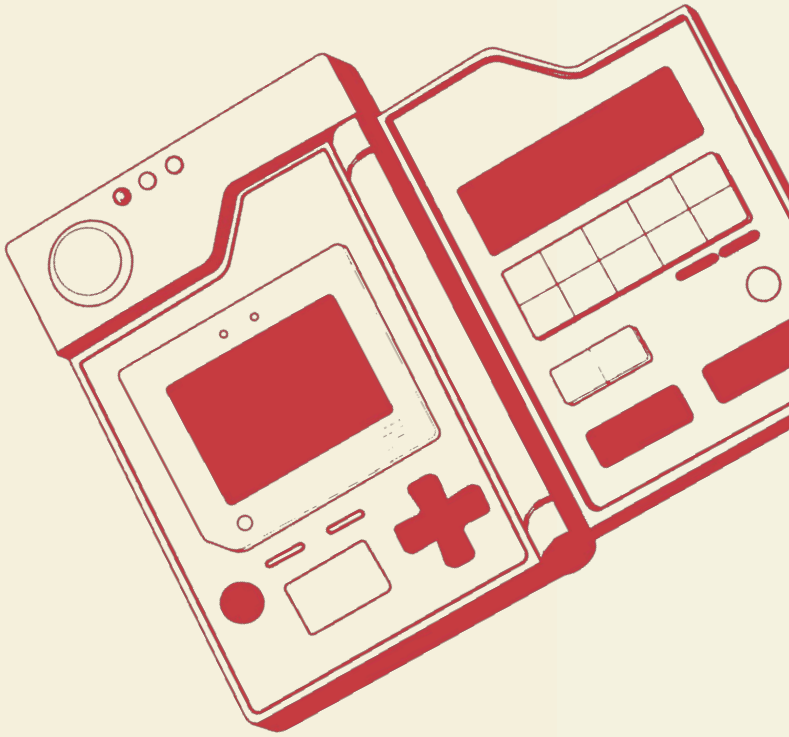
Our original idea for the project was to build a smart Pokédex using an ESP32-CAM. The plan was for the camera to capture an image of a Pokémon, connect to Google search or image recognition tools, identify the word or name that appeared most often, and use that to determine which Pokémon had been scanned. Once identified, the system would find the corresponding Pokédex number and use the PokéAPI to fetch data such as the Pokémon's name, type, stats, and Pokédex entry, which would then be displayed on the device. Although this concept was exciting because it used live image recognition and online data retrieval, we felt that it relied too heavily on external APIs and automation rather than reflecting the kind of hands-on input-output logic and physical interaction we had been learning in ODT.







Original Pokedex



ORIGINAL CONCEPT ART

Our final design reimagined the Pokédex as a self-contained physical system rooted in embedded electronics and tangible interaction. Rather than relying on image recognition, we created a set of Pokémon figures using paper cutouts and foldable forms, each with a unique resistor embedded inside. These resistance values are read through an ADC circuit, where the measured ADC value allows the system to identify which Pokémon is connected to the scan platform, the figures plug in via a male jack into a female jack, creating a secure and repeatable connection. When the power button is pressed, the system initiates a boot sequence in which the NeoPixels light up in series while the OLED display plays a startup animation and shows "POKÉDEX." The user then places a Pokémon figure on the platform, plugs it in, and presses the scan button, prompting the system to read the ADC value and call the corresponding Pokémon function. The NeoPixel lights up in the colour associated with that Pokémon's type, the OLED displays its name, type, and stats, and a pre-recorded audio clip reads out its Pokédex entry aloud. Once a Pokémon has been scanned and registered, the Pokéball becomes active. Each Pokémon has a hardcoded catch rate, and triggering the Pokéball runs a random function against this value to determine the outcome. A shaking animation simulates the catch sequence, and a light indicator, along with the OLED display, reveals whether the Pokémon was successfully caught or managed to escape. The system can be reset at any time by long-pressing the power button. This version of the project stayed true to the spirit of what we had learned, prioritising physical making, coded logic, and a layered interactive experience built from the ground up.

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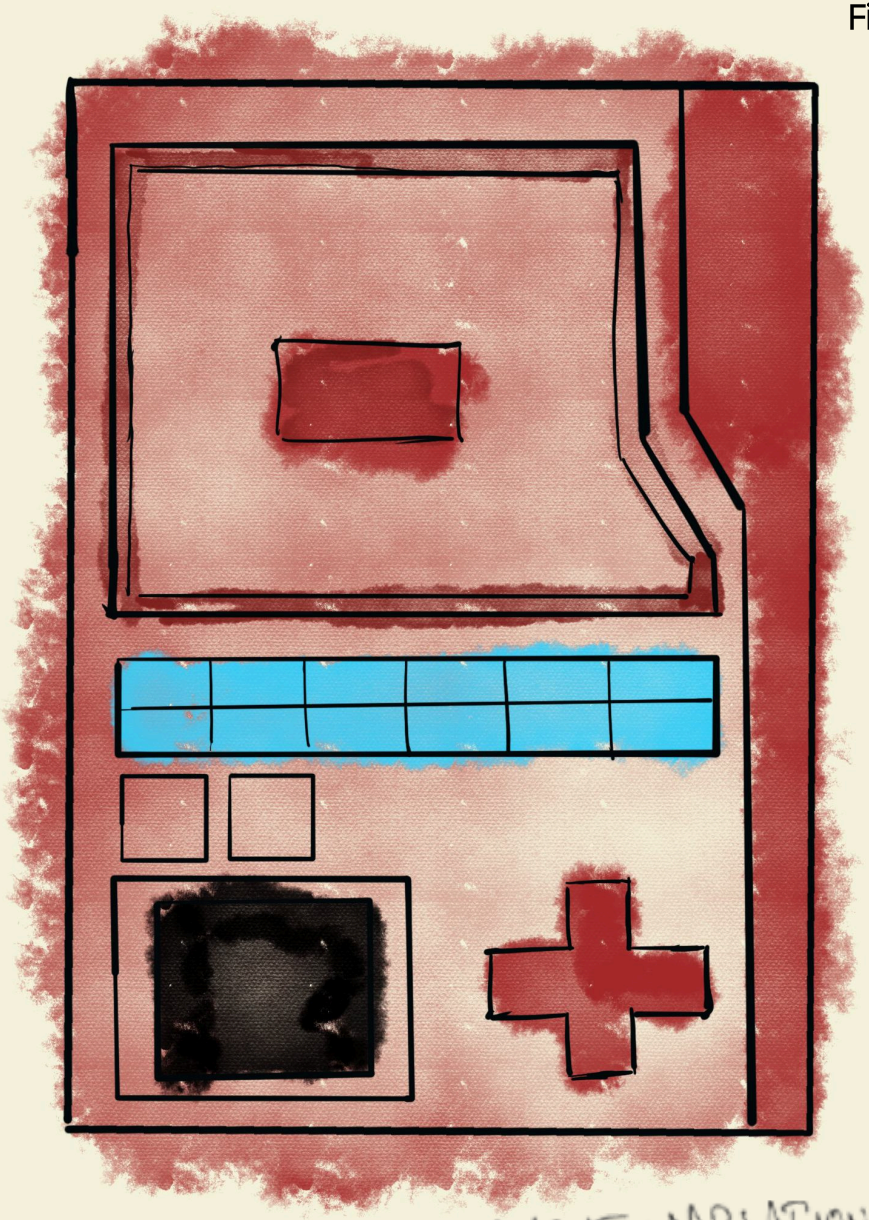
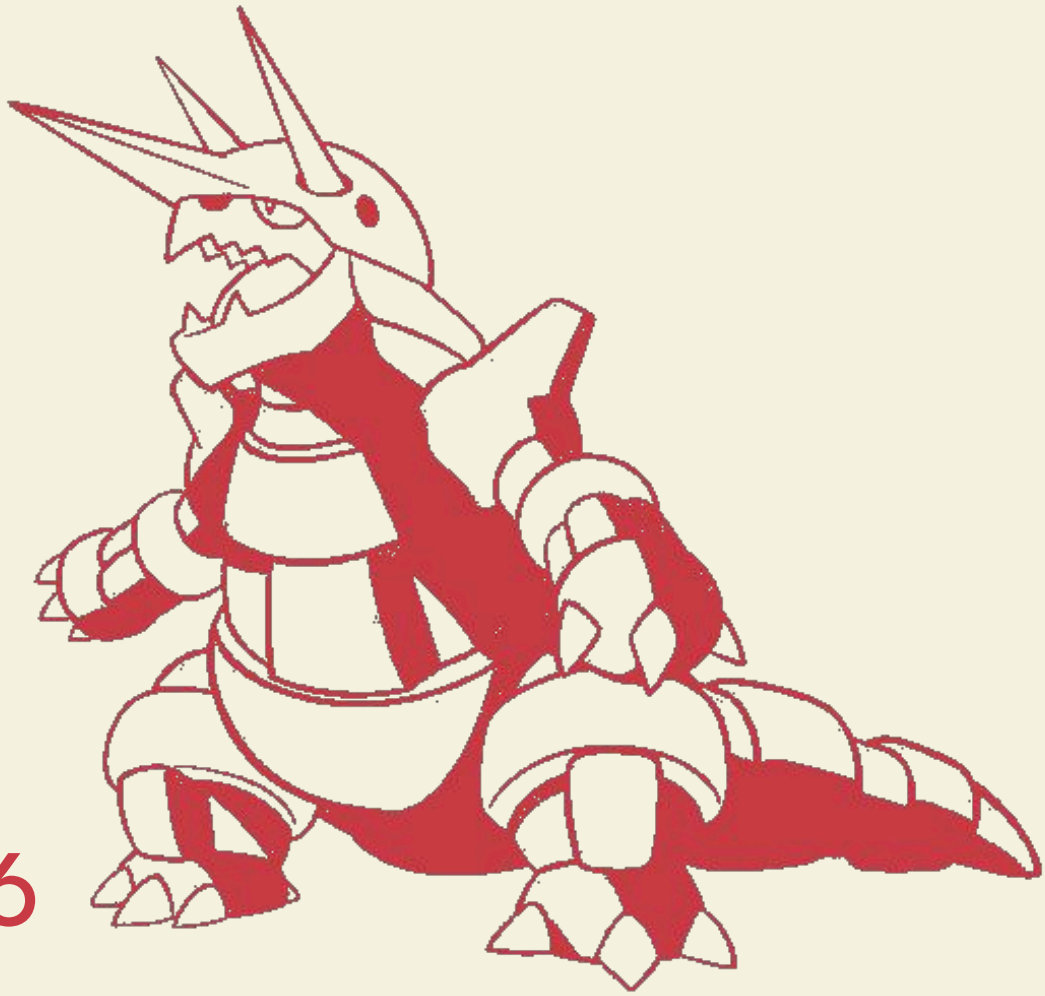
Final layout



FINAL CONCEPT ART

WHY WE CHOSE THIS

We chose to work on this because we wanted to build something that was genuinely code intensive at its core , where the complexity lives in the logic of identification and sorting rather than in physical mechanics. Our first ODT summative touched on a similar idea but used capacitive touch as a fairly rough input method, so this felt like a natural opportunity to develop a more refined and considered version of that concept. We also made a deliberate choice to move away from large moving elements, motors are finicky to work with, and more importantly, most of our peers were already heading in that direction. As longtime Pokémon fans, the theme chose itself, it gave the project a strong conceptual identity while letting us build something technically ambitious and personally meaningful.



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